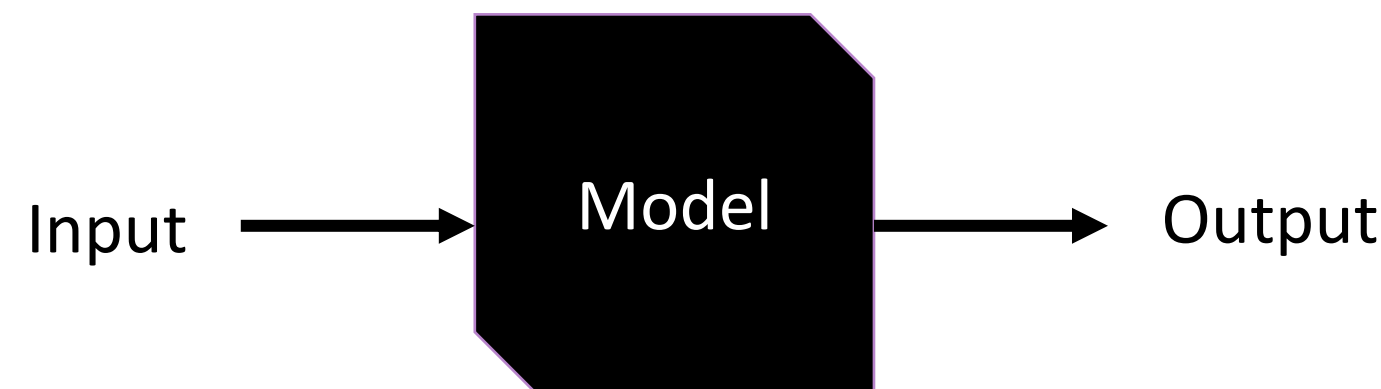
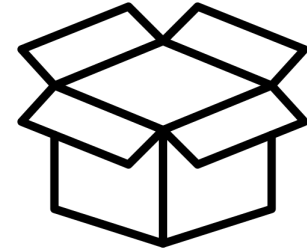


Introduction

- Rapid advancements in **deep learning** have significantly transformed **Natural Language Processing (NLP)**, enabling **language models (LMs)** to achieve impressive performance across a wide range of tasks.
- However, increasing **model complexity** often compromises **interpretability**, turning these models into **black boxes** that obscure the reasoning behind their decisions.



- Explainable Artificial Intelligence (XAI)** seeks to address this challenge by providing interpretable explanations, which are essential for **trust**, **accountability**, and **usability** among diverse stakeholders



- Most XAI research is **Anglo-Centric**; explanation methods have been developed and evaluated in the context of English NLP, with limited attention to Arabic.

- This creates a gap in understanding how effective these methods are when applied to **Arabic language models**.

- To address this, our study evaluates explanation techniques on Arabic language models across two dimensions:

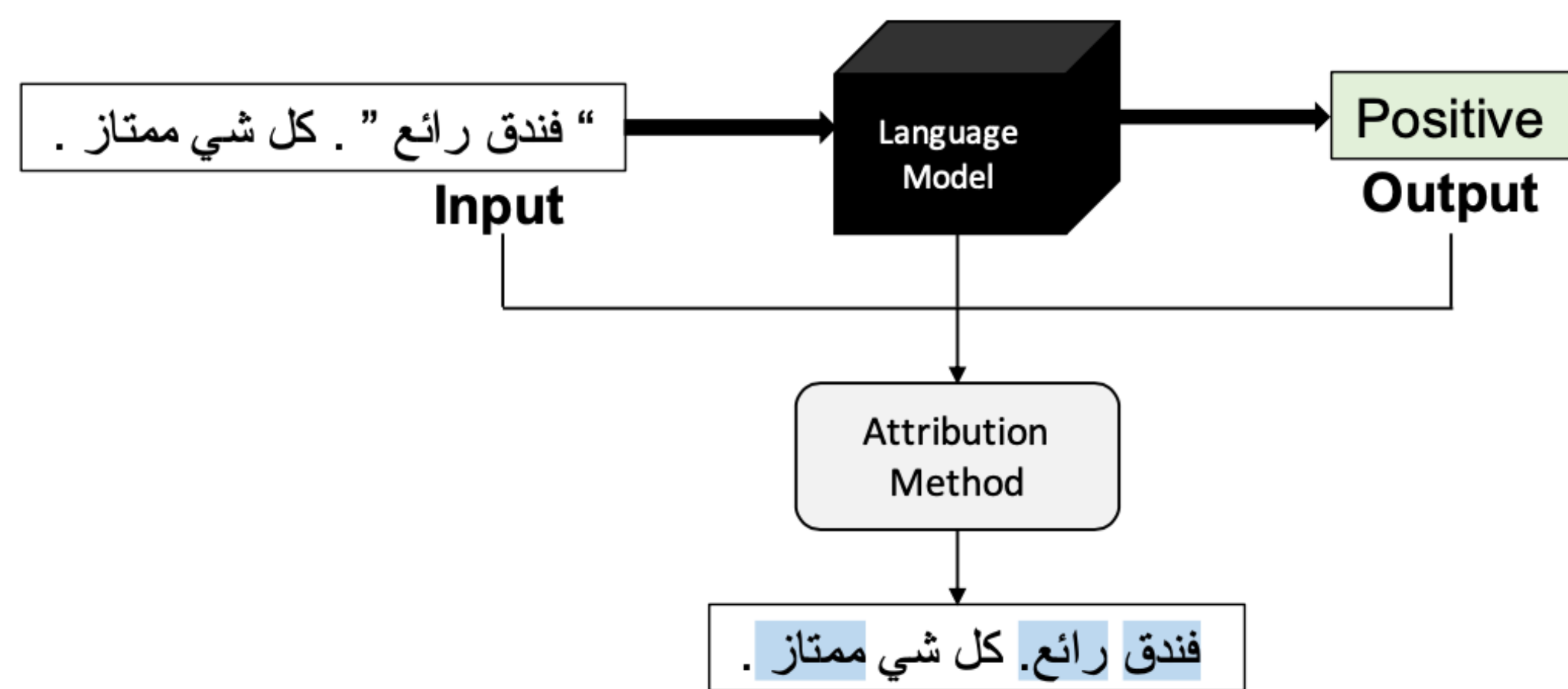
- Faithfulness** – the degree to which explanations reflect the model's actual reasoning.
- Plausibility** – how well explanations align with human expectations and reasoning.

Literature Review

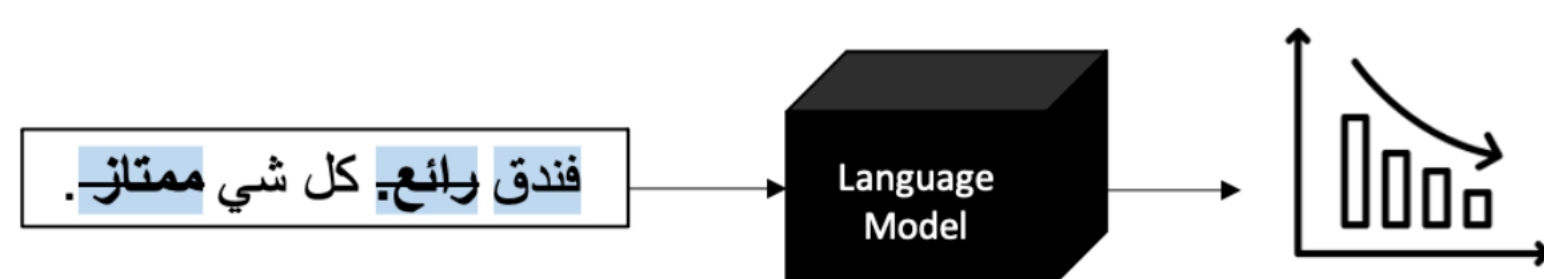
- ✓ **XAI in NLP.** The most widely used techniques include *local feature attribution methods* (e.g., gradient-, perturbation-based approaches).
- ✓ **Evaluation of Explanations.** No ground truth exists for “ideal” explanations. Evaluations typically assess **faithfulness** to the model's reasoning and **plausibility** to human judgment [2].
- ✓ **Limited XAI Work for Arabic Models.** Few studies have explored explainability in Arabic NLP, mostly applying LIME to sentiment analysis tasks [3, 4]. To our knowledge, this is the first systematic evaluation of multiple XAI methods across tasks in Arabic LMs.

Methods

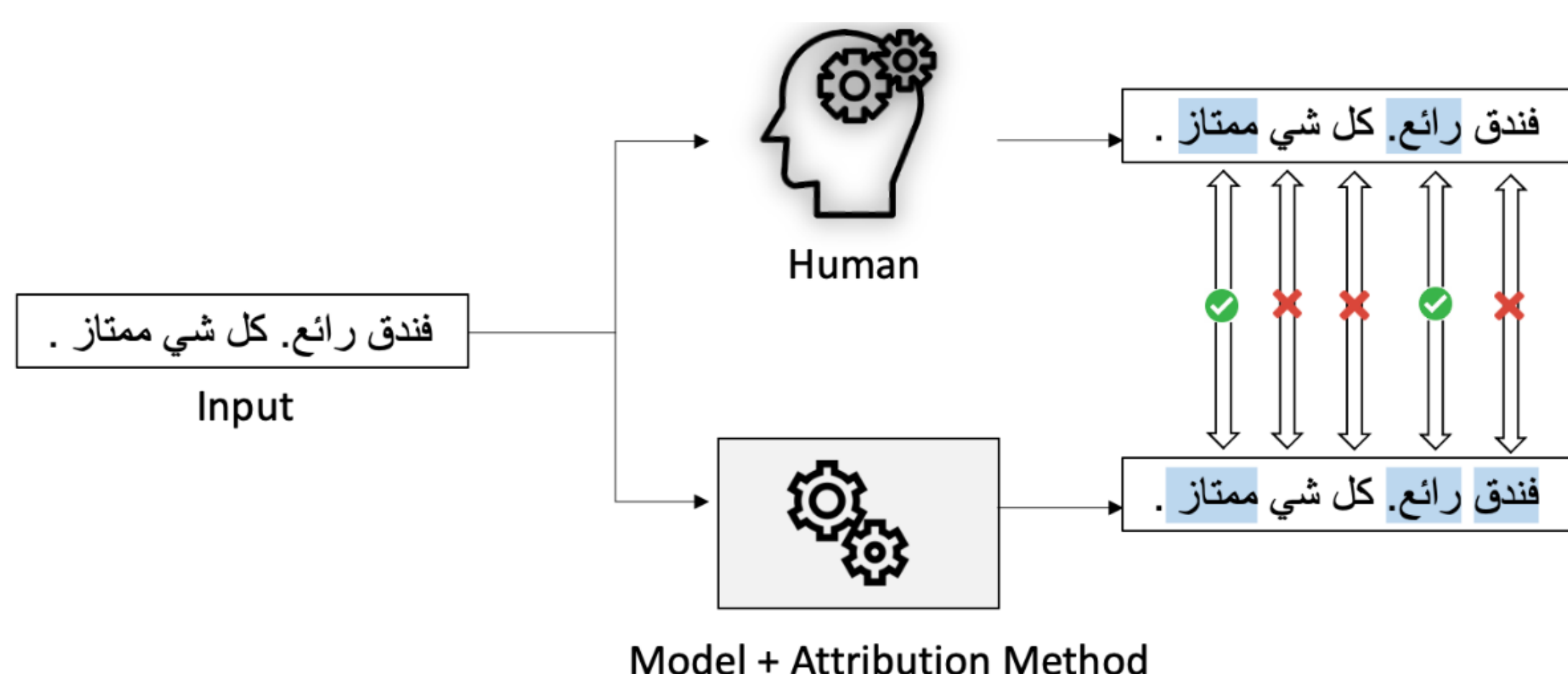
Attribution method



Faithfulness



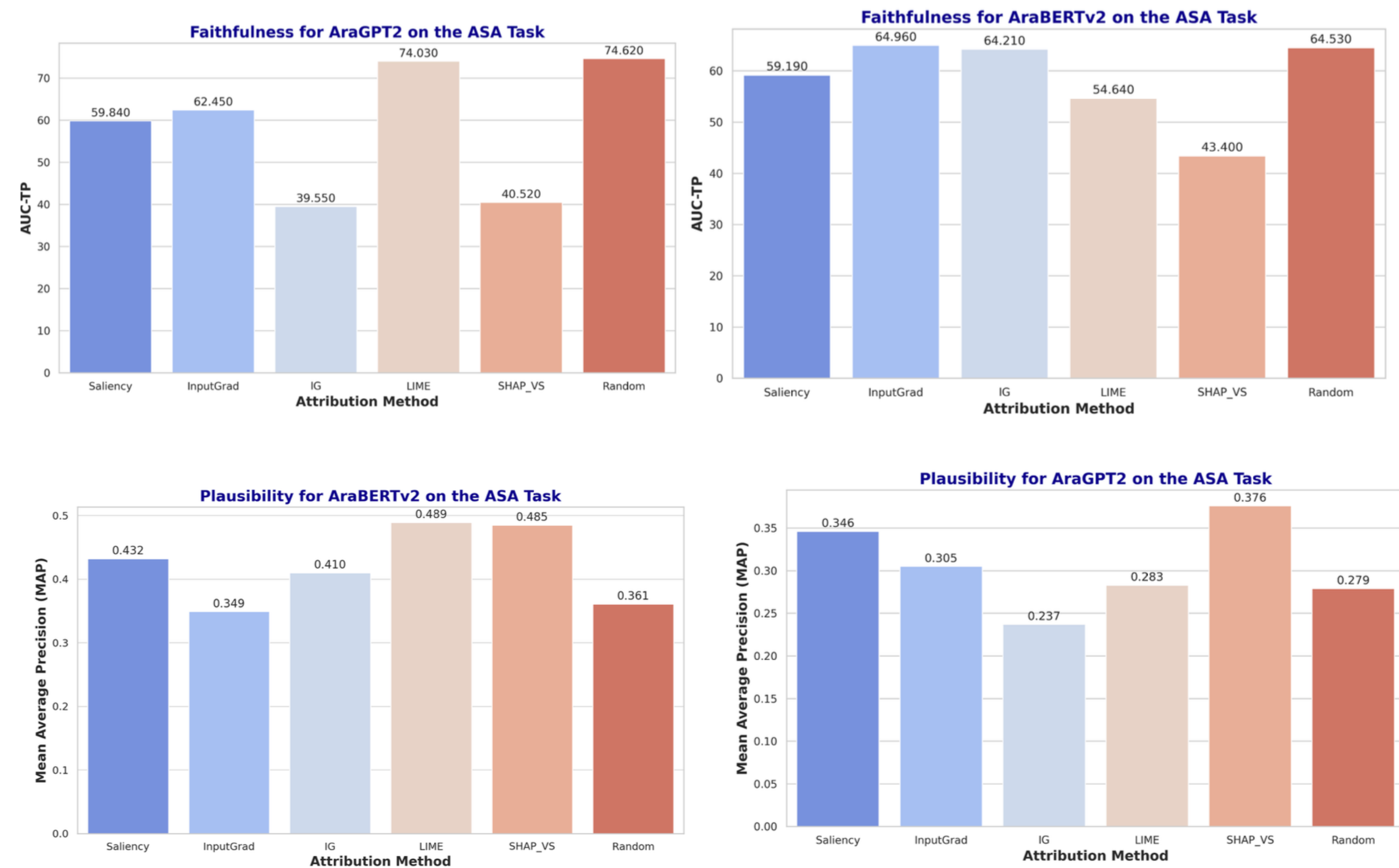
Plausibility



Experimental Setup

- Sentiment Analysis
Hotel Arabic-Reviews Dataset (HARD)
- Semantic Question Similarity
Mawdoo3 Q2Q dataset (MQ2Q)
- Models: AraBERT-v2, AraGPT2
- Attribution Methods
 - Gradient-based: Saliency, InputXGradient, Integrated Gradients
 - Perturbation-based: LIME, Shapley Value Sampling.
- Faithfulness: Area Under the Threshold-Performance Curve.
- Plausibility: Mean Average Precision.

Results & Discussion



- SHAP_VS, a **perturbation-based** method, shows the highest alignment with human rationales.
- Gradient-based methods** were more effective in reflecting the model's internal reasoning, indicating higher faithfulness.
- The findings reinforce that **faithfulness and plausibility are fundamentally distinct** and should not be used interchangeably.
- The variation in top-performing methods across different tasks suggests that the effectiveness of explanation techniques is both **model- and task-dependent**.

Conclusion

- We conducted a systematic evaluation of five explanations methods on different Arabic LMs across different tasks.
- Gradient-based methods are generally more faithful, while perturbation-based methods better align with human plausibility.
- Future Work.** Extend to more tasks and models, and explore additional explainability methods and evaluation metrics for both faithfulness and plausibility.

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